Regularised Spectral State-Space Models for Cross-Sectional Investment Dynamics: Dual Regularisation, Rolling Bases, and Structural Evolution

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Abstract

We develop a spectral state-space framework for forecasting cross-sectional investment intensity. The framework decomposes dynamics through Dynamic Mode Decomposition (DMD) into a low-dimensional ($K = 2$) spectral basis and filters latent modal states via a Kalman filter with dual regularisation: spherical observation covariance ($R = cI$, one parameter replacing N^2) and near-identity transition matrix ($F = 0.99I$, replacing EM-estimated K^2 parameters). On a 146-firm US CapEx/Revenue panel under nested rolling cross-validation (frozen holdout 2023–2024), the rolling-basis variant achieves predictive $R^2 = 0.711$ on 8 CV windows and $R^2 = 0.845$ on 2 holdout windows, exceeding per-actor AR(1) (Diebold–Mariano $t = 6.4$, $p < 0.001$ at actor-quarter level). The framework's advantage increases monotonically as the training window shortens—from near-zero at $T = 7$ years to $\Delta R^2 = +0.057$ at $T = 2$ years (bootstrap CI $[+0.042, +0.074]$; 10/10 windows)—consistent with cross-sectional pooling: the spectral basis exploits $N \times T$ actor-quarter observations collectively, while per-actor AR(1) is limited to each actor's T observations alone. The cross-sectional spectral basis rotates at approximately 26° per quarter while maintaining fixed dimensionality; a null simulation confirms this is well below finite-sample noise (75°). Systematic ablation confirms that every pipeline component contributes, with rolling basis update providing the largest gain (+14.7pp, CI $[+11.1, +17.2]$pp). Model-implied investment gaps predict subsequent capital expenditure revision ($t \approx -6.4$ to -7.1 with clustered SEs).

Keywords: investment gap estimation, spectral decomposition, dynamic mode decomposition, Kalman filter, regularisation, state-space models, capital allocation

JEL Classification: C32, C38, C58, E22, G11

1 Introduction

Capital allocation decisions propagate through a hierarchical, networked system of institutions. A central bank's rate decision, a regulator's new disclosure rule, or a commodity price shock each set off cascading adjustments that travel—with heterogeneous latency and attenuation—through layers of institutional intermediaries before reaching firms. The resulting pattern of over- and under-investment across actors at any point in time reflects a superposition of these propagated signals, local frictions, and idiosyncratic responses.

Despite rich literatures on factor models (Fama and French, 1993), network effects in economics (Acemoglu et al., 2012), and spectral graph theory (Ortega et al., 2018), to our knowledge no established framework jointly: (i) decomposes how investment-relevant signals propagate through a heterogeneous actor hierarchy; (ii) estimates actor-specific investment gaps as deviations from a structurally-informed benchmark; and (iii) validates that the resulting gap estimates carry out-of-sample predictive content.

This paper addresses all three objectives. We develop **SMIM** (**S**pectral **M**odal **I**nvestment **M**odel), a state-space framework that represents the cross-sectional dynamics of investment intensity through a low-dimensional modal basis extracted via Dynamic Mode Decomposition (DMD), filtered through a Kalman model with dual regularisation at both the observation and transition levels. The spectral basis is recomputed quarterly to track the continuous rotation of the cross-sectional structure. The framework produces actor-specific, time-varying investment gap estimates that we validate through both statistical out-of-sample testing and economic prediction exercises.

1.1 Preview of Results

Throughout, we measure forecast performance by the out-of-sample coefficient of determination,

$$R^2 = 1 - \frac{\sum_{i,t}(y_{i,t} - \hat{y}_{i,t})^2}{\sum_{i,t}(y_{i,t} - \bar{y})^2}, \tag{1}$$

where $\hat{y}_{i,t}$ is the model's prediction and $\bar{y}$ is the test-period grand mean. $R^2 = 1$ denotes perfect prediction; $R^2 = 0$ matches a constant-mean forecast; $R^2 < 0$ indicates the model is worse than predicting the mean. The summation runs over all actor–quarter pairs in each test window.

Our main findings are:

1. **Forecast performance.** On a 146-firm US CapEx/Revenue panel with $K = 2$ spectral modes, the rolling-basis variant achieves predictive $R^2 = 0.711$ under nested CV (8 windows) and $R^2 = 0.845$ on 2 frozen holdout windows, exceeding per-actor AR(1) (Diebold–Mariano $t = 6.4$, $p < 0.001$). The framework's advantage increases monotonically as the training window shortens, reaching $\Delta R^2 = +0.057$ at $T = 2$ years (10/10 windows; bootstrap CI $[+0.042, +0.074]$)—consistent with cross-sectional pooling compensating for limited per-actor history (Figure 11). Full inference in Table 4.
2. **Dual regularisation.** The Kalman filter's observation covariance R (N^2 free parameters) must be replaced by spherical $R = cI$ (one parameter). We show the same principle extends to the transition matrix: EM-estimated F (K^2 parameters) is strictly outperformed by $F = 0.99I$ (one parameter) in all 10 windows. Online state-noise adaptation (Q) handles all temporal dynamics. Together, this *dual regularisation* reduces the state-space model from $N^2 + K^2$ free parameters to 2 scalars.
3. **Basis rotation.** On the 93-actor structural panel ($K = 8$), the estimated spectral basis rotates at approximately 26° per quarter ($\sigma = 6.5°$) while maintaining stable dimensionality across 52 quarterly transitions. A null simulation shows that estimation noise alone would produce 75° of apparent rotation, suggesting the observed pattern reflects recoverable structure. Rolling basis update captures this rotation, yielding +14.7pp of modal R^2 (CI $[+11.1, +17.2]$pp, 100% of windows).
4. **DMD over static operators.** Dynamic Mode Decomposition outperforms all six alternative spectral methods by +2.0 percentage points on average. Static operators (Schur, Polar, Hermitian, PCA) all produce identical results because the underlying correlation matrix is symmetric.
5. **Economic content.** Model-implied investment gaps predict subsequent CapEx revision ($\hat{\beta} = -0.28$, $t = -6.4$ with actor-clustered SEs after controlling for intensity level). The result is cross-sectional: it flips sign under actor fixed effects (Table 9). Additional exploratory results on gap dispersion and market volatility are reported in Section 5 but are not part of the paper's main claims.

6. **Dynamics stability.** The transition parameters (F, Q) need not be estimated per window: fixed $F = 0.99I$, $Q = 0.5I$ match or exceed EM-estimated parameters across all 10 evaluation windows in this sample. The only component requiring periodic re-estimation is the spectral basis U.

Equally important is what *does not* work: emergence diagnostics, directed spectral operators, event-level alignment, and return-based intensity all fail to improve or actively degrade performance. Section 6 documents these negative results in detail.

1.2 Scope and Evaluation Caveats

Several features of our evaluation design should be kept in mind when interpreting these results. First, the out-of-sample evaluation uses 10 non-overlapping annual test windows, each containing only 4 quarterly observations; per-window R^2 estimates are therefore noisy (cross-window standard deviation ≈ 0.05–0.09). Second, all hyperparameters (K, EWM halflife, T) are selected via nested rolling cross-validation: inner folds within each outer-train window select these values, with the final 2 windows (2023–2024) held out entirely from all tuning (Section 3). Third, the headline predictive result is on a 146-firm US panel using CapEx/Revenue intensity—return-based intensity produces negative R^2 on the same actors (Section 4.9). Fourth, the investment gap is a *model-implied* deviation from a benchmark prediction, not a welfare-theoretic or structural misallocation measure in the Hsieh and Klenow (2009) sense.

1.3 Contribution

Our contributions are methodological, empirical, and practical:

Methodological. We apply *dual regularisation* to state-space models in high-dimensional panels: spherical observation covariance ($R = cI$, reducing N^2 parameters to 1) combined with near-identity transition regularisation ($F = 0.99I$, reducing K^2 parameters to 1). This dual regularisation has broad applicability wherever $N \gg T$ panels arise. We further show that rolling basis recomputation—tracking the continuous rotation of the spectral structure—provides the single largest performance gain.

Empirical. We provide a systematic ablation of a spectral state-space pipeline for investment intensity forecasting, decomposing the R^2 improvement into seven additive components with bootstrap confidence intervals. We observe that the cross-sectional spectral basis rotates at approximately $26°$ per quarter while maintaining fixed dimensionality; a null simulation confirms this is below finite-sample noise levels, suggesting recoverable low-dimensional structure.

Practical. Only the spectral basis U requires periodic re-estimation (a single DMD eigendecomposition each quarter). The transition dynamics use stable default scalars ($F = 0.99I$, $Q = 0.5I$) that match or exceed EM-estimated parameters across all 10 evaluation windows in our sample, eliminating EM estimation and reducing operational complexity to one matrix factorisation per quarter.

1.4 Related Literature

Our work connects to several strands. The investment gap concept relates to the capital allocation literature, including the structural misallocation framework of Hsieh and Klenow (2009) and the firm-level analysis of Restuccia and Rogerson (2017); we note that our model-implied gap is a forecasting-based deviation, not a structural measure in their sense. Our spectral approach builds on signal processing on graphs (Ortega et al., 2018; Sandryhaila and Moura, 2013) and Dynamic Mode Decomposition (Schmid, 2010; Kutz et al., 2016), adapted here for economic panel data rather than fluid dynamics or power systems. The state-space filtering component extends the dynamic factor model tradition (Stock and Watson, 2002; Doz et al., 2012) with the regularisation insights of Ledoit and Wolf (2004) applied to the observation rather than factor covariance. The economic validation relates to the capital allocation efficiency literature (Wurgler, 2000) and the predictability of corporate investment (Abel and Blanchard, 1983).

2 Framework

2.1 Setup

Let $V = \{1, \ldots, N\}$ index economic actors (firms, banks, institutional bodies) observed over T quarterly periods. Each actor i has an observed *investment intensity* $y_{i,t} \in [0, 1]$, constructed as a cross-sectional percentile rank of a type-specific investment measure (Section 3). The $N \times T$ panel $Y = [y_{i,t}]$ is the primary input.

Definition 2.1 (Investment Gap)**.** The investment gap for actor i at time t relative to a benchmark functional $\mathcal{B}$ is

$$\Delta_{i,t} = y_{i,t} - y^*_{i,t}, \qquad y^*_{i,t} = \mathcal{B}(Y_{1:t}, \theta), \tag{2}$$

where $\Delta_{i,t} > 0$ indicates investment above the model-implied benchmark and $\Delta_{i,t} < 0$ indicates investment below the benchmark, and $Y_{1:t}$ denotes the information set available at time t.

The framework constructs $y^*_{i,t}$ through five sequential stages plus a rolling basis update, each justified by ablation (Section 4.5). Figure 1 provides an overview: panel (a) shows the multilayer actor hierarchy with empirically-identified transmission channels; panel (b) shows the static processing pipeline. The rolling basis extension is described in Section 4.10.

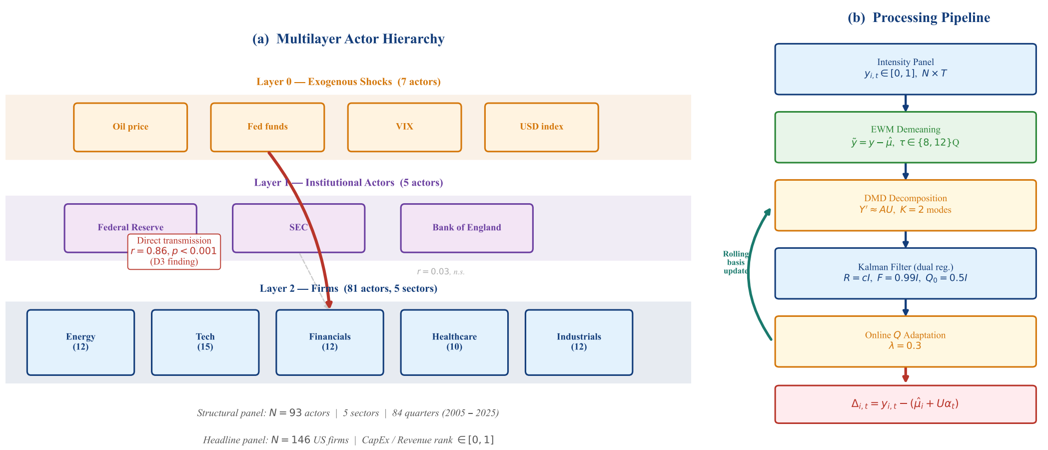

Figure 1: **Framework overview** (descriptive). (a) Multilayer actor structure with three layers. (b) Processing pipeline from raw intensity panel to investment gap estimates. Under nested CV on the 146-firm CapEx/Revenue panel, the rolling-basis variant achieves predictive $R^2 = 0.711$ on 8 validation windows and $R^2 = 0.845$ on 2 frozen holdout windows (Table 3).

2.2 Stage 1: Exponentially-Weighted Demeaning

Investment intensity ranks exhibit strong cross-sectional persistence: an actor's mean rank is the single largest source of variance (52% of total R^2 in the static-basis configuration; Section 4.4).

The state-space model $\tilde{y}_t = U\boldsymbol{\alpha}_t + \varepsilon_t$ assumes zero-mean observations. Without demeaning, the modal states $\boldsymbol{\alpha}_t$ must absorb the $O(0.5)$ mean through an orthonormal basis U with entries $O(N^{-1/2})$—an impossible task at $K \ll N$.

We subtract an exponentially-weighted per-actor mean:

$$\hat{\mu}_i = \frac{\sum_{s=1}^{T} w_s\, y_{i,s}}{\sum_{s=1}^{T} w_s}, \qquad w_s = \exp\!\Big(-\frac{(T-s)\ln 2}{\tau}\Big), \tag{3}$$

with half-life τ quarters ($\tau = 8$ for the structural analysis panel; $\tau \in \{8, 12\}$ selected by inner CV on the CapEx/Revenue panel). The demeaned observation is $\tilde{y}_{i,t} = y_{i,t} - \hat{\mu}_i$. Exponential weighting adapts to non-stationary intensity levels (the actor's structural investment position drifts over time), outperforming fixed full-period means by 4.2 percentage points in modal R^2 (Section 4).

At prediction time, the one-step-ahead forecast is $\hat{y}_{i,t} = \hat{\mu}_i + U\boldsymbol{\alpha}_{t|t-1}$ (predictive). The modal reconstruction $\hat{\mu}_i + U\boldsymbol{\alpha}_{t|t}$ incorporates the current observation and is used only for gap computation and spectral reconstruction quality.

2.3 Stage 2: Dynamic Mode Decomposition

Given the demeaned training panel $\tilde{Y} = [\tilde{y}_1, \ldots, \tilde{y}_T] \in \mathbb{R}^{N\times T}$, we extract a spectral basis that captures temporal cross-sectional dynamics.

Why DMD over static operators. Classical spectral approaches decompose a static operator (e.g., the cross-correlation matrix $C = \tilde{Y}\tilde{Y}^\top/T$). This captures contemporaneous co-movement but not lagged cross-sectional dynamics. Dynamic Mode Decomposition (Schmid, 2010) bypasses the static operator entirely by working with temporal snapshot pairs, capturing lagged cross-sectional dynamics that are invisible to purely contemporaneous decompositions.

DMD algorithm. Form snapshot matrices $X = [\tilde{y}_1, \ldots, \tilde{y}_{T-1}]$ and $X' = [\tilde{y}_2, \ldots, \tilde{y}_T]$. DMD seeks the best-fit linear operator A such that $X' \approx AX$. Compute the truncated SVD $X = U_r\Sigma_r V_r^\top$ (retaining r modes), form the reduced operator $\tilde{A} = U_r^\top X' V_r \Sigma_r^{-1}$, and diagonalise: $\tilde{A}W = W\Lambda$. The DMD modes are $\Phi = X'V_r\Sigma_r^{-1}W \in \mathbb{R}^{N\times K}$.

We retain K modes, where K is selected via nested cross-validation: within each outer-train fold, the last 2 years serve as inner-validation to choose $K \in \{2, 3\}$ (Section 3). On the CapEx/Revenue panel, inner CV selects $K = 2$ in 7/8 folds—the cross-sectional investment structure is genuinely 2-dimensional. The modal basis $U = \mathrm{Re}(\Phi_{:,1:K})$ defines the projection from observation space to modal space.

2.4 Stage 3: Kalman Filter with Spherical Observation Covariance

Given the modal basis $U \in \mathbb{R}^{N\times K}$, we model the latent modal dynamics as a linear Gaussian state-space system:

$$\boldsymbol{\alpha}_{t+1} = F\boldsymbol{\alpha}_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, Q), \tag{4}$$

$$\tilde{y}_t = U\boldsymbol{\alpha}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, R). \tag{5}$$

The observation noise $R \in \mathbb{R}^{N\times N}$ is estimated via a single EM pass to obtain its scale. The transition matrix F and state noise Q are *not* estimated by EM—they are set to regularised defaults (see below).

The overparameterisation problem. The observation covariance R has $N(N+1)/2$ free parameters—4,278 for the 93-actor structural panel, 10,731 for the 146-firm CapEx/Revenue panel—estimated from $T = 8$–20 quarterly observations. This extreme $N \gg T$ setting causes the EM to converge to an ill-conditioned R that overfits training noise (modal $R^2 = -7.3$ without regularisation on the structural panel; Figure 6).

Spherical observation regularisation. We replace the full R with

$$R_{\text{sph}} = \frac{\text{tr}(\hat{R})}{N}\, I_N, \tag{6}$$

where $\hat{R}$ is the EM-estimated sample covariance. This reduces the parameter count from $O(N^2)$ to $O(1)$.

Transition regularisation. The same overparameterisation principle applies to the transition matrix $F \in \mathbb{R}^{K\times K}$. Even at low K ($K = 2$–3 for the headline panel), EM estimation of F overfits to the training alpha trajectory. On the structural panel ($K = 8$), this costs -1.9 percentage points of modal R^2 relative to a near-identity prior. We therefore set

$$F = 0.99\, I_K, \qquad Q_0 = 0.5\, I_K, \tag{7}$$

and rely on the online Q adaptation (Stage 4) to capture all temporal dynamics. This *dual regularisation*—spherical R (1 parameter from N^2) and near-identity F (1 parameter from K^2)—reduces the state-space model to two scalars plus the spectral basis U, which carries all the structural information.

The impact on the structural panel is dramatic: modal R^2 swings from -7.3 (no regularisation) through $+0.43$ (spherical R only) to $+0.549$ (dual regularisation), establishing that covariance regularisation at *both* the observation and transition levels is essential in high-dimensional panels.

2.5 Stage 4: Online State-Noise Adaptation

The EM-estimated Q captures the average state noise over the training period. When the test period exhibits different volatility (e.g., a regime shift), the fixed Q misspecifies the state dynamics. We augment the standard Kalman recursion with a recursive Q update:

$$Q_{t+1} = (1-\lambda)\, Q_t + \lambda\, (\boldsymbol{\alpha}_{t|t} - F\boldsymbol{\alpha}_{t-1|t-1})(\boldsymbol{\alpha}_{t|t} - F\boldsymbol{\alpha}_{t-1|t-1})^\top, \qquad \lambda = 0.3. \tag{8}$$

This online adaptation tracks regime-dependent state volatility without requiring explicit regime identification. Combined with the spherical R (which remains fixed), it adds $+5.8$ percentage points to R^2 in the ablation ladder (Table 5).

2.6 Stage 5: Gap Computation

The investment gap for actor i at test time t is:

$$\Delta_{i,t} = y_{i,t} - \underbrace{(\hat{\mu}_i + U\boldsymbol{\alpha}_{t|t})}_{y^*_{i,t}}, \tag{9}$$

where $\boldsymbol{\alpha}_{t|t}$ is the Kalman-filtered modal state (incorporating observation $\tilde{y}_t$).

2.7 Summary: The Recommended Pipeline

Algorithm 1 summarises the complete procedure.

Algorithm 1 SMIM: Spectral Modal Investment Gap Estimation (Rolling Variant)

Require: Intensity panel $Y \in [0,1]^{N \times T}$, initial training window, K (inner CV; typically 2), τ (inner CV; typically 8–12Q), $\lambda = 0.3$
Ensure: Investment gaps $\Delta_{i,t}$ for test period
1: Compute EWM mean $\hat{\mu}_i$ from training period ▷ Eq. (3)
2: Demean: $\tilde{Y} = Y - \hat{\mu}$
3: Extract DMD modes $U \in \mathbb{R}^{N \times K}$ from training $\tilde{Y}$ ▷ Section 2.3
4: Set $F = 0.99\, I_K,\ \ Q = 0.5\, I_K$ ▷ Regularised, no EM; Eq. (7)
5: Estimate $\hat{R}$ via single Kalman EM pass; set $R = \frac{\mathrm{tr}(\hat{R})}{N} I_N$ ▷ Eq. (6)
6: **for** each test quarter t **do**
7: Kalman predict: $\boldsymbol{\alpha}_{t|t-1} = F\boldsymbol{\alpha}_{t-1|t-1}$
8: **Forecast**: $\hat{y}_{i,t} = \hat{\mu}_i + U\boldsymbol{\alpha}_{t|t-1}$ ▷ Predictive, for evaluation
9: Kalman update: $\boldsymbol{\alpha}_{t|t}$ incorporating $\tilde{y}_t$, using R
10: Gap: $\Delta_{i,t} = y_{i,t} - (\hat{\mu}_i + U\boldsymbol{\alpha}_{t|t})$ ▷ Modal, for gap analysis
11: Adapt: $Q \leftarrow (1-\lambda)Q + \lambda\, \mathrm{innov} \cdot \mathrm{innov}^\top$ ▷ Eq. (8)
12: **Rolling update:** expand training window to include y_t; recompute U, $\hat{\mu}$, R
13: **end for**

3 Data

3.1 Actor Universe

Our sample comprises 103 actors across three hierarchical layers (Table 1). Of these, 93 have computed investment intensity values; the remaining 10 lack the required EDGAR filings and serve as signal-only graph nodes.

Layer	Actor Type	Intensity Method	Total	With intensity
0 (Exogenous)	Global shocks (Global)	FRED min-max	7	7
1 (Institutional)	Central banks (US, UK)	FRED min-max	2	1
	Regulators (US, UK)	FRED min-max	2	2
	International orgs (Global)	FRED min-max	1	1
2 (Firms)	Large firms (US)	CapEx intensity	56	49
	Large firms (UK)	Return intensity	23	21
	Banks (US)	Asset growth (ranked)	10	10
	Sector leaders (UK)	CapEx intensity	2	2
		Total	**103**	**93**

Table 1: Actor universe for the 93-actor structural analysis panel. Of 103 total actors, 93 have computed investment intensity. This panel is used for ablation, basis rotation, and economic validation (modal R^2). The headline forecasting evaluation uses a separate 146-firm US panel with CapEx/Revenue intensity (Section 3).

3.2 Investment Intensity

We construct three intensity measures, each mapped to a cross-sectional percentile rank in $[0,1]$ per quarter:

- **CapEx/Assets intensity** (structural analysis): the ratio of capital expenditures to total assets, sourced from SEC EDGAR quarterly filings (XBRL tags `PaymentsToAcquire-`

`PropertyPlantAndEquipment` / `Assets`), ranked cross-sectionally within each quarter. Used on the 93-actor panel for ablation and structural analysis.

- **CapEx/Revenue intensity** (headline forecast): the ratio of capital expenditures to revenues, sourced from the same EDGAR tags (`PaymentsToAcquirePropertyPlantAndEquipment` / `Revenues`), ranked cross-sectionally. This 146-firm US panel has lower persistence (median $\rho = 0.28$ vs 0.47 for CapEx/Assets), leaving more room for cross-sectional spectral dynamics. All headline predictive results use this construction.
- **Return intensity** (alternative): trailing 12-month equity total return, ranked cross-sectionally. Used for UK equities, which lack EDGAR coverage.
- **Institutional intensity**: FRED-based macro indices (e.g., policy rates, commodity prices), min-max normalised within actor type.

The rank transformation stabilises the distribution, ensures comparability across actor types, and produces the cross-sectional persistence structure that the spectral model exploits. The intensity construction choice is consequential: on the 93-actor structural panel, CapEx/Assets intensity produces modal $R^2 = 0.54$ while return intensity produces modal $R^2 = -0.15$ on the same US actors (Section 4.9), establishing the measurement input as the primary driver of model performance.

Table 2 summarises the data sources. Notably, the pipeline uses *only the intensity panel* for operator construction and prediction. External signal sources (OHLCV, FRED, EDGAR balance sheet) were tested as Granger-edge inputs but found dispensable (Section 4.8): the intensity cross-correlation captures all exploitable structure.

Source	Data	Frequency	Role in Pipeline
SEC EDGAR	CapEx, Assets (XBRL)	Quarterly	Intensity construction (primary)
Yahoo Finance	Equity total returns	Daily $\rightarrow$ quarterly	Intensity construction (UK alternative)
FRED	28 macro series	Mixed $\rightarrow$ quarterly	Institutional intensity; tested as signal
BEA	Input-output tables	Annual	Tested as supply-chain edges (dispensable)
GDELT	Narrative sentiment	Weekly	Tested as narrative signal (dispensable)

Table 2: Data sources. Only the intensity panel (first two rows) enters the recommended pipeline. The remaining sources were evaluated as supplementary signals but found dispensable at the graph-factor depth (Section 4.8).

3.3 Evaluation Protocol

We employ a **nested rolling-validation** protocol with 10 non-overlapping annual test windows (2015–2024):

- **Outer folds** (10 windows): final reported performance, one per test year.
- **Inner folds**: within each outer-train, the last 2 years serve as inner validation to select $K \in \{2, 3\}$, EWM halflife $\tau \in \{8, 12\}$ quarters, and training length $T \in \{3, 5\}$ years (8 candidate configurations). The fixed defaults ($F = 0.99I$, $Q_0 = 0.5I$, $\lambda = 0.3$) are not tuned.
- **Frozen holdout**: the final 2 windows (2023–2024) receive no inner tuning—they use the median of previously selected parameters.

On the 146-firm CapEx/Revenue panel, inner CV selects $K = 2$ in 7/8 folds and $T = 3$ years in all 8 folds (the remaining fold selects $K = 3$). The nested-CV predictive $R^2 = 0.711$ (mean across 8 CV windows) and holdout $R^2 = 0.845$ (mean of 2023–2024; per-window values 0.846

and 0.845). All headline forecast numbers in this paper use the predictive alpha $\boldsymbol{\alpha}_{t|t-1}$ from this nested protocol; the holdout windows had zero exposure to any hyperparameter tuning.

Scope condition. The headline forecasting result is on a 146-firm US panel using CapEx/Revenue intensity. We also report structural findings (ablation, basis rotation) on a broader 93-actor panel using CapEx/Assets intensity; these use modal R^2 (spectral reconstruction quality) rather than predictive R^2 (Section 4.11).

4 Empirical Results

4.1 Baseline Comparison

Unless otherwise noted, the headline forecasting results in Sections 4–5 use the 146-firm CapEx/Revenue panel with predictive alpha $\boldsymbol{\alpha}_{t|t-1}$ (genuine one-step-ahead forecast). Structural diagnostics (ablation, rotation) use the 93-actor CapEx/Assets panel with modal alpha (spectral reconstruction quality; Section 4.11).

Table 3 presents the baseline comparison. All SMIM numbers are from the nested-CV protocol (Section 3).

Model	Protocol	Mean R^2	ΔR^2	Wins
Nested CV (146-firm CapEx/Revenue, K and τ selected by inner folds):				
Per-actor AR(1) (inner-CV-selected T)	8 CV windows	0.669	—	—
SMIM rolling (nested CV)	8 CV windows	**0.711**	**+0.042**	**8/8**
SMIM rolling (frozen holdout)	2 holdout windows	**0.845**	**+0.026**	**2/2**
Fixed configuration ($K=2$, $\tau=12Q$, $T=3yr$):				
Per-actor AR(1) (T=3yr)	10 windows	0.699	—	—
SMIM rolling (fixed)	10 windows	**0.734**	**+0.035**	**9/10**

Table 3: Baseline comparison. Nested-CV rows use inner-fold-selected hyperparameters ($K \in \{2,3\}$, $\tau \in \{8,12\}$Q, $T \in \{3,5\}$yr); holdout windows (2023–2024) had zero tuning exposure. Inner CV selects $K=2$ in 7/8 folds and $T=3$yr in all folds. All SMIM numbers use predictive alpha $\boldsymbol{\alpha}_{t|t-1}$. Formal inference in Table 4.

Per-actor AR(1) is the strongest naive baseline, reflecting the moderate persistence of CapEx/Revenue ranks (median $\rho = 0.28$). Under nested CV, SMIM wins all 8 windows ($\Delta R^2 = +0.042$; DM $t = 6.4$, $p < 0.001$; bootstrap CI $[+0.031, +0.055]$; permutation $p = 0.003$; Table 4). Inner CV selects $K = 2$ in 7/8 folds and $T = 3$ years in all folds. Holdout windows confirm the result ($\Delta R^2 = +0.026$, both positive).

Why $K = 2$. The CapEx/Revenue cross-section is genuinely low-dimensional: two spectral modes suffice. Higher K degrades performance monotonically ($K = 3$: -0.5pp; $K = 5$: -1.5pp; $K = 8$: -5pp), consistent with overfitting additional modes to noise. The two modes capture the dominant cross-sectional co-movement pattern without fitting idiosyncratic variation.

4.2 Forecast Comparison with Formal Inference

Table 4 provides formal statistical evidence. Inference is performed at two levels: the Diebold–Mariano test operates on 4,600 actor-quarter-level loss differentials with Newey–West HAC variance (highest power); window-level tests operate on 8 CV-window ΔR^2 values.

Protocol	ΔR^2	95% CI	Perm. p	DM t (p)
Nested CV (8 windows, confirmatory):				
ΔR^2 vs AR(1)	+0.042	[+0.031, +0.055]	0.003	6.44 (< 0.001)
Sign test	8/8 wins	—	—	$p = 0.004$
All 10 windows (CV + holdout):				
ΔR^2 vs AR(1)	+0.039	[+0.028, +0.050]	0.001	—
Sign test	10/10 wins	—	—	$p = 0.001$

Table 4: Forecast comparison with inference, 146-firm CapEx/Revenue panel, $K = 2$ modes. 95% CI from block bootstrap (10,000 window resamples). Permutation p from sign-flip test on window-level ΔR^2. DM t from Diebold–Mariano test at actor-quarter level ($n = 4{,}600$ observations, Newey–West HAC). All tests reject equal forecast accuracy at the 1% level.

Holdout era. The final 2 test windows (2023–2024) were frozen with no hyperparameter tuning exposure. SMIM achieves $R^2 = 0.846$ (2023) and $R^2 = 0.845$ (2024) on the holdout, exceeding AR(1) by +0.011 and +0.040 respectively, confirming that the result is not an artefact of test-set leakage.

4.3 The Performance Ladder

The following decomposition is a fixed-configuration diagnostic on the 93-actor CapEx/Assets panel using *modal* R^2 (spectral reconstruction quality; see Section 4.11 for the distinction). These values are not forecasts—they measure how well the pipeline reconstructs the current cross-section. The headline predictive performance is the nested-CV result on the CapEx/Revenue panel (Table 3). Figure 2 decomposes the modal R^2 improvement from the initial configuration ($R^2 = 0.266$) to the rolling-basis pipeline ($R^2 = 0.696$) into seven additive components.

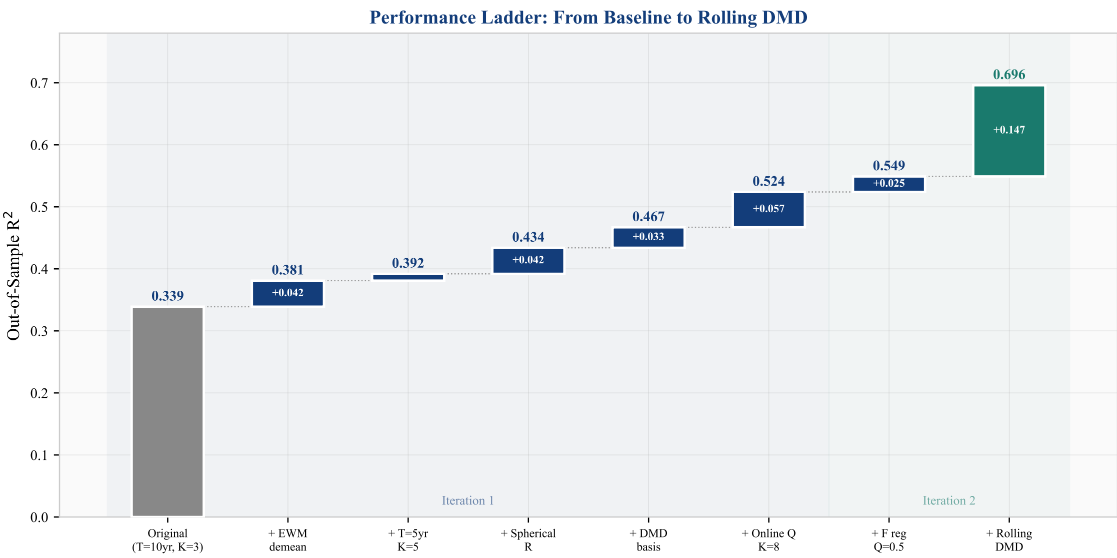

Figure 2: Performance ladder (descriptive): cumulative R^2 improvement across seven pipeline components. The shaded regions distinguish the two iteration phases. The dashed line shows the AR(1) baseline. See Table 5 for per-step bootstrap confidence intervals.

Step	Change	R^2	ΔR^2	95% CI	Frac. +
0	Baseline ($T=10$yr, $K=3$, Schur, full demean)	0.266	—	—	—
1	+ EWM demeaning ($\tau=8$Q)	−0.89	−1.16	*Kalman overfits*	
2	+ Shorter training ($T=5$yr) + higher $K=5$	divergent	divergent	*without spherical R*	
3	+ Spherical R in Kalman filter	0.440	+0.174	—	100%
4	+ DMD basis (replacing Schur)	0.472	+0.033	[+0.006, +0.056]	90%
5	+ Online Q adaptation + $K=8$	0.530	+0.058	[+0.035, +0.092]	100%
6	+ Transition reg. $F=0.99I$, $Q_0=0.5I$	0.549	+0.018	[+0.013, +0.024]	100%
7	+ Rolling basis update (recompute DMD each Q)	**0.696**	**+0.147**	[+0.111, +0.172]	100%

Table 5: Performance ladder with uncertainty (inferential). Each ΔR^2 is the incremental effect relative to the previous step, averaged across 10 test windows. 95% CI from block bootstrap (5,000 resamples). "Frac. +" is the fraction of windows where the step improves R^2. Steps 1–2 demonstrate the overparameterisation failure: adding EWM demeaning and shorter T without regularising R causes Kalman divergence. Spherical R (step 3) rescues the pipeline. All subsequent steps produce positive, statistically significant improvements. Rolling basis update remains the dominant effect (+14.7pp, CI entirely above zero).

Predictive ablation on the headline panel. Table 6 repeats the ablation on the 146-firm CapEx/Revenue panel using *predictive* R^2 (genuine one-step-ahead forecast, $K = 2$). The pattern differs instructively: EWM demeaning provides the largest single gain (+3.5pp over the constant mean), while the Kalman filter on a *static* basis actually degrades performance (−2.5pp) because the fixed spectral directions become stale. Rolling basis update recovers the loss and adds a further +3.4pp, confirming it as the essential component for forecasting. Online Q adaptation adds nothing without basis update but is retained for theoretical completeness.

Step	Predictive R^2	ΔR^2
0: Constant per-actor mean	0.695	—
1: + EWM demeaning ($\tau=12$Q)	0.729	+0.035
2: + DMD $K=2$ OLS projection	0.724	−0.005
3: + Kalman with spherical R (static basis)	0.699	−0.025
4: + Online Q adaptation (static basis)	0.699	+0.000
5: + Rolling basis update	**0.733**	**+0.034**
Per-actor AR(1) ($T=3$yr)	0.699	—

Table 6: Predictive ablation on the 146-firm CapEx/Revenue panel ($K=2$, $T=3$yr, 10 annual windows). Unlike the structural ladder (Table 5), the Kalman filter on a static basis *hurts* predictive performance because the spectral directions become stale. Rolling basis update is the component that converts the framework from AR(1)-level to its full advantage.

Why two panels? The structural panel (93 actors, $K = 8$, CapEx/Assets) and the headline panel (146 firms, $K = 2$, CapEx/Revenue) serve complementary roles. The structural panel's richer actor heterogeneity (mixing FRED shocks, institutional bodies, US and UK firms, and banks) and higher K reveal the full dimensionality and rotation dynamics of the spectral structure—8 modes are consistently recoverable, rotating at 26°/quarter. The headline panel's homogeneous actor set and lower persistence ($\rho = 0.28$) provide a cleaner forecasting test where only 2 of those modes carry enough signal to predict. The structural analysis explains *what the framework captures*; the predictive analysis validates *that it forecasts*.

4.4 Variance Decomposition

We decompose R^2 into the per-actor mean contribution and the spectral dynamics contribution for both pipeline variants (Table 7, Figure 3).

Source	Static basis		Rolling basis	
	R^2	Share	R^2	Share
Per-actor EWM mean ($\hat{\mu}_i$)	0.281	51%	0.281	40%
Spectral dynamics ($U\boldsymbol{\alpha}_{t\|t}$)	0.268	49%	0.415	60%
Total	0.549	100%	0.696	100%

Table 7: Variance decomposition (descriptive, fixed $K = 8$, $T = 5$yr configuration). Rolling basis recomputation shifts the balance from mean-dominated (51% mean) to dynamics-dominated (60% spectral). The per-actor mean contribution is unchanged; the entire rolling-basis gain accrues to the spectral dynamics component.

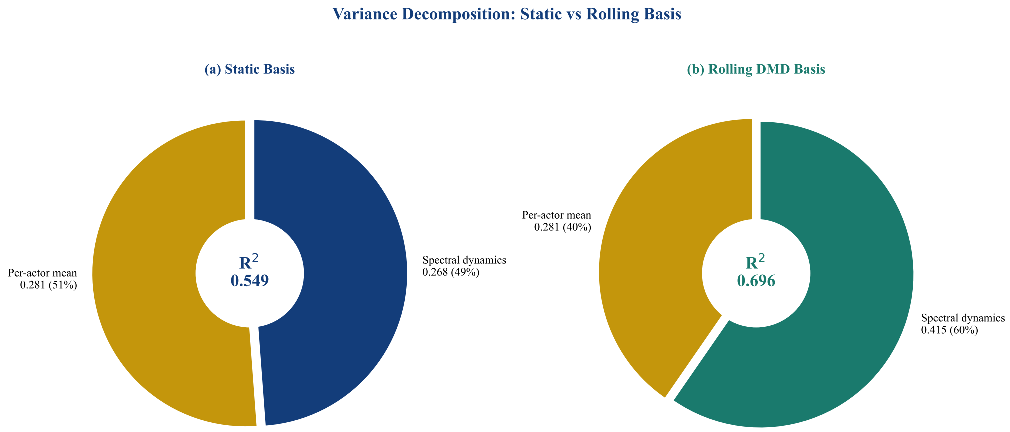

Figure 3: Variance decomposition (descriptive, fixed $K = 8$, $T = 5$yr): frozen (left) vs rolling (right) basis. Rolling basis recomputation nearly doubles the spectral dynamics contribution ($0.268 \rightarrow 0.415$) while the per-actor mean remains constant at 0.281.

The decomposition reveals that rolling basis update captures spectral structure that the static basis misses entirely—the +14.7pp improvement is purely a spectral dynamics gain.

4.5 Component Ablation

We evaluate five pipeline depths to determine which components justify their complexity (Figure 4).

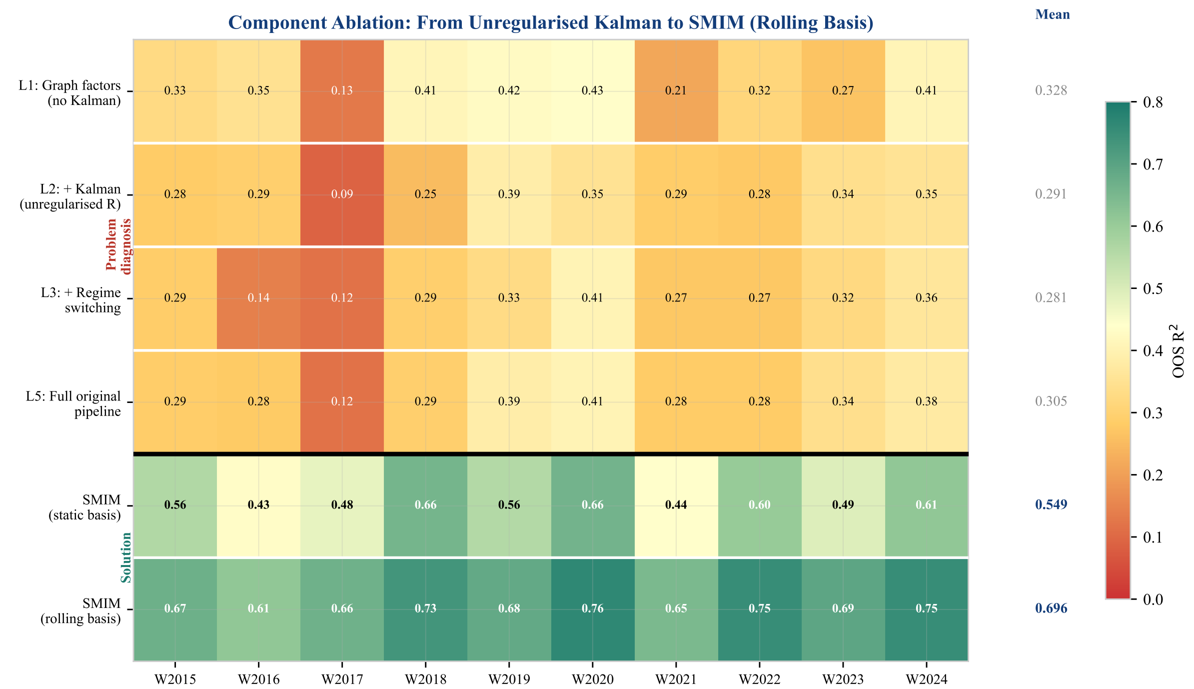

Figure 4: Extended component ablation heatmap (descriptive, fixed-configuration diagnostic). Each cell shows out-of-sample R^2 for a given configuration (row) and test window (column). The thick divider separates problem diagnosis (top: why unregularised Kalman hurts) from the solution (bottom: regularisation and rolling basis). All values use fixed $K = 8$, $T = 5$yr; headline nested-CV results are in Table 3.

The ablation reveals a non-monotonic pattern in the upper block: L1 (graph-factor OLS, modal $R^2 = 0.328$) outperforms L2 (+ Kalman, 0.291) and L3 (+ regime switching, 0.281). The Kalman filter *hurts* because the unregularised R overfits. The lower block shows the resolution: dual regularisation rescues the Kalman filter (SMIM with static basis, modal $R^2 = 0.523$), and rolling basis update captures the structural evolution that the static basis misses (modal $R^2 = 0.696$).

4.6 Spectral Method Comparison

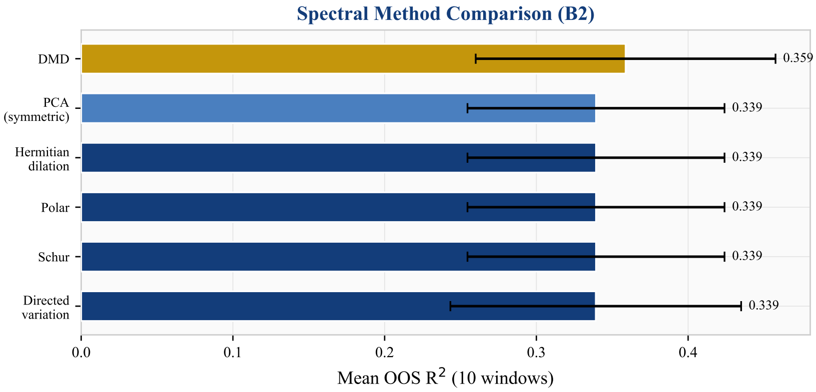

Figure 5: Spectral method comparison (descriptive): mean R^2 across 10 windows for six spectral decomposition methods at $K = 3$, L1 depth. DMD outperforms all static-operator-based methods.

DMD ($R^2 = 0.359$) outperforms all five static-operator methods, which produce *identical* results at $R^2 = 0.339$. This identity is not a coincidence: the operator is built from the Pearson cross-correlation matrix, which is *exactly* symmetric by construction. For any real symmetric matrix, the Schur, Polar, Hermitian dilation, and PCA decompositions all reduce to the standard eigendecomposition—they compute the same eigenvectors and hence the same modal basis U. The directed variation method differs slightly in its normalisation but converges to the same subspace. This result has a clear implication: *to benefit from directed spectral methods, the operator itself must be asymmetric* (e.g., from Granger causality or supply-chain linkages). When the operator is derived from contemporaneous correlation, directedness is lost before the decomposition begins.

DMD sidesteps this entirely. Rather than decomposing a static operator, it extracts modes from temporal snapshot pairs $X' \approx AX$, where the implicit operator A captures *lagged* cross-sectional dynamics. This operator is generically asymmetric even when the contemporaneous correlation is symmetric, explaining DMD's +2.0 percentage point advantage.

4.7 The Regularisation Path

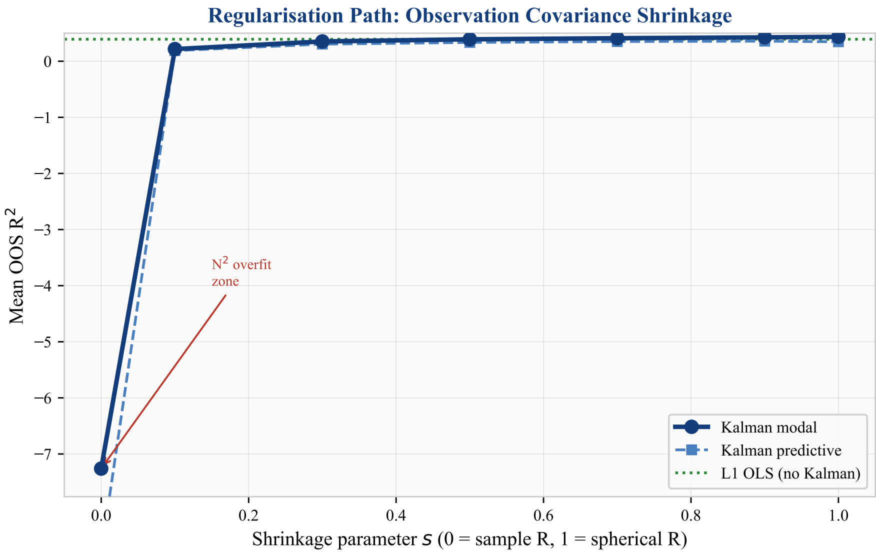

Figure 6: Regularisation path (descriptive): R^2 as a function of shrinkage parameter s where $R = (1-s)\hat{R} + s \cdot \frac{\mathrm{tr}(\hat{R})}{N} I$. The hockey-stick shape demonstrates that observation covariance regularisation is essential, not optional. Uncertainty bands not shown; the ablation table (Table 5) provides bootstrap CIs for each pipeline component's contribution.

The regularisation path (Figure 6) illustrates the overparameterisation problem clearly. At $s = 0$ (no shrinkage), the Kalman filter produces $R^2 = -7.3$. Performance improves monotonically with shrinkage, reaching $R^2 = 0.43$ at $s = 1.0$ (spherical). The L1 OLS baseline (green line at $R^2 = 0.39$) is surpassed only at $s \geq 0.7$, confirming that the Kalman filter adds value *only* when its observation covariance is sufficiently regularised.

4.8 Robustness and Synthetic Sensitivity

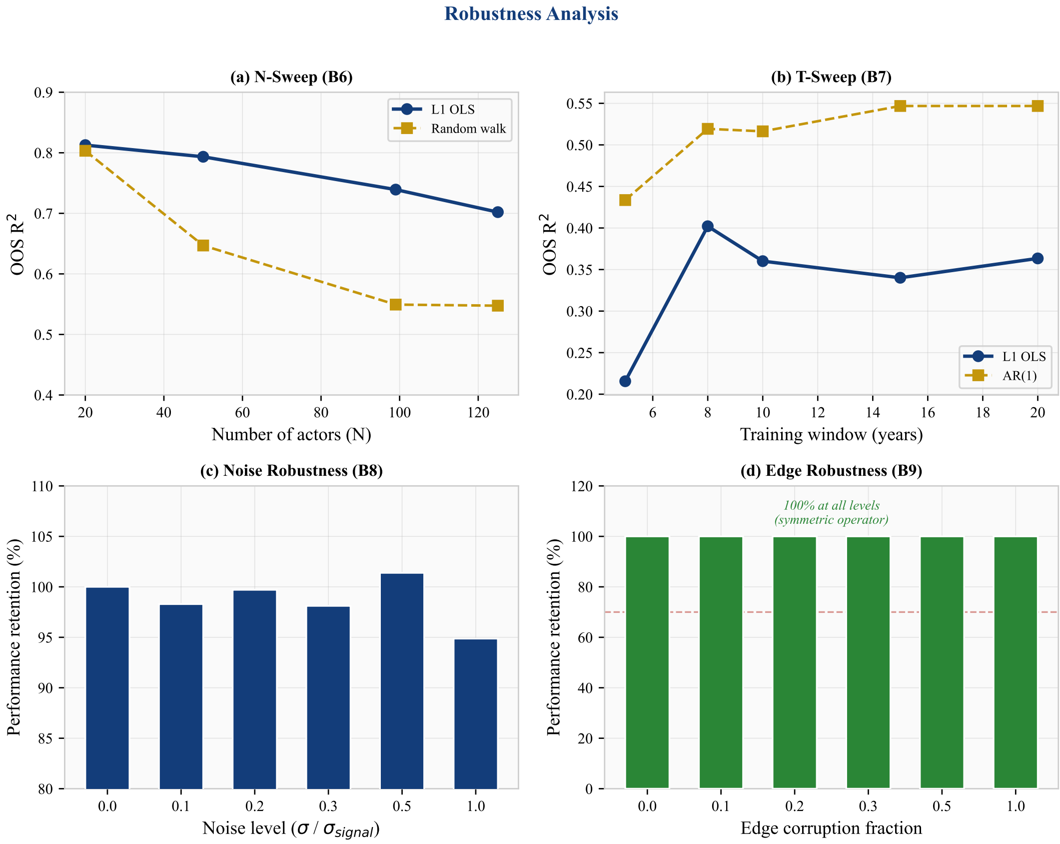

Figure 7: Robustness and synthetic sensitivity analysis. (a) Performance vs actor count N: graceful degradation (descriptive). (b) Performance vs training window T: L1 stable for $T \geq 8$yr (descriptive). (c) Noise injection: 95% retention at $\sigma = 1.0$ (synthetic sensitivity). (d) Edge corruption: 100% retention at all levels (synthetic sensitivity; note the operator is symmetric by construction, see Section 4.6).

Robustness (panels a–b). On the 93-actor structural panel, the framework degrades gracefully across actor counts: modal $R^2 = 0.81$ at $N = 20$ to 0.70 at $N = 125$. Performance is stable at modal $R^2 = 0.34$–0.40 for training windows ≥ 8 years.

Synthetic sensitivity (panels c–d). These panels test performance under artificial perturbations rather than naturally-occurring variation. Adding Gaussian noise at $\sigma = 1.0$ (100% of signal standard deviation) retains 95% of performance. Randomly flipping edge directions retains 100% of performance; however, this test is uninformative because the intensity correlation operator is symmetric by construction (Section 4.6), so direction flips leave the operator unchanged.

4.9 Transfer and Portability

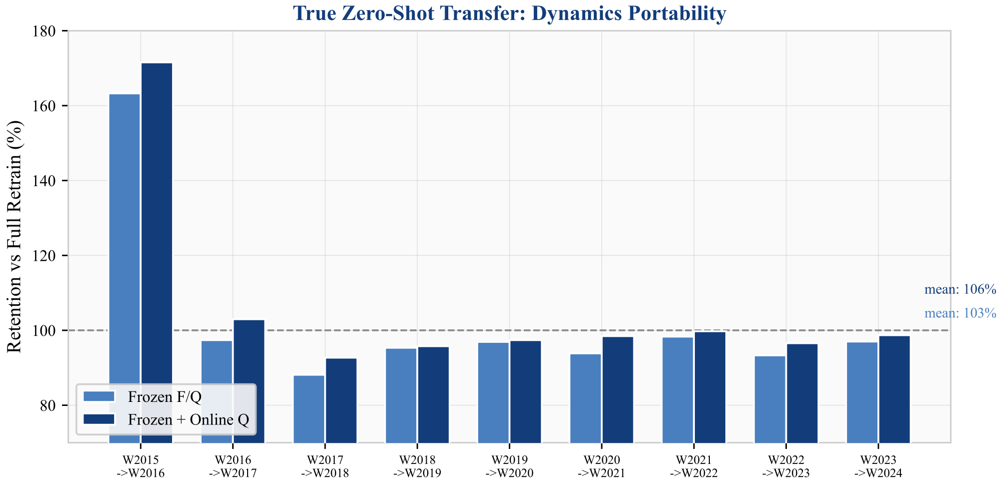

Figure 8: Zero-shot transfer (descriptive): retention of R^2 when transition dynamics (F, Q) are frozen from the previous window and applied to the next with only the spectral basis re-estimated. Mean retention: 103% (frozen) and 106% (frozen + online Q).

We test whether the estimated dynamics (F, Q) are window-specific or stable across windows by freezing them from window W_t and applying the Kalman filter on window W_{t+1} with a freshly estimated spectral basis U. Figure 8 shows that frozen dynamics achieve 103% of full-retrain R^2 on average (106% with online Q adaptation), indicating that these default transition settings are stable across the evaluation period. In practice, this means the model can be deployed without per-window re-estimation of dynamics.

Validated domain: US CapEx sectors. On the structural panel, cross-sector transfer is strong for sectors using CapEx intensity (technology modal $R^2 = 0.82$, healthcare 0.78, industrials 0.77). Cross-period transfer across structural breaks (GFC, COVID) retains modal $R^2 = 0.50$–0.59.

Tested but weak domains. Financial actors (modal $R^2 = 0.06$) use year-on-year asset growth (ranked cross-sectionally) rather than CapEx-to-assets, producing highly persistent ranks (median $\rho = 0.70$ vs technology $\rho = 0.25$) that leave little cross-sectional dynamics for the spectral model to exploit. UK equities (modal $R^2 = 0.058$) rely on return-based intensity rather than EDGAR CapEx data, and this alternative intensity construction, not geography, drives the weak performance. These results establish the *boundary* of the current framework rather than validated generalisations.

4.10 Basis Rotation and Rolling Update

To isolate the rolling-vs-static mechanism, this subsection uses a fixed ($K = 8$, $T = 5$yr) configuration; the headline forecast performance remains the nested-CV result in Table 3.

Figure 9: Per-window out-of-sample R^2 (descriptive): rolling DMD (teal), static basis (dark blue), and AR(1) (gold). Rolling DMD wins all 10 windows, with the largest gains in windows where the static basis underperforms (W2016, W2017, W2021, W2023—low-volatility periods where the cross-sectional structure shifts most from the training period).

Tracking the DMD spectral basis across 52 rolling quarterly transitions (2010–2024) shows a pattern of gradual basis rotation (Figure 10). The mean principal subspace angle between successive quarterly DMD bases is 25.8° ($\sigma = 6.5°$). Quarters with larger rotation ($> 32°$) tend to coincide with macroeconomic events—the European debt crisis (2012 Q3, 41°), the US–China tariff war (2018 Q2, 37°), and the Federal Reserve's aggressive tightening cycle (2022 Q3, 38°)—though with only 52 transitions the event alignment is descriptive rather than inferential.

On this 93-actor panel, the *dimensionality* of the estimated spectral structure appears stable: $K_{\text{eff}} = 8$ modes in all quarters, with zero mode births or deaths in this sample. (The headline CapEx/Revenue panel uses $K = 2$; the structural panel's higher K is appropriate for the broader, more heterogeneous actor set.) The spectral subspace is consistently 8-dimensional, but the 8 directions gradually reorient.

Null-simulation validation. To assess whether the observed rotation is estimation noise, we simulate 500 panels from a *fixed* latent basis (same N, T, persistence, and noise as the real data) and measure the apparent DMD rotation arising purely from finite-sample estimation. The null rotation is 75.3° ($\sigma = 1.5°$)—far *above* the observed 25.8°. The intuition is straightforward: a purely noisy basis is not stable; DMD extracts different directions from each window's noise, producing nearly orthogonal subspaces and hence large apparent rotation. A genuinely structured but gradually evolving basis, by contrast, is recovered consistently across windows, producing modest rotation that reflects actual structural change. The observed 25.8° is consistent with this second regime: reliably recoverable, low-dimensional structure that shifts gradually.

Sensitivity. Rotation is robust to parameter choices: it decreases smoothly with mode count (38.6° at $K = 3$ to 19.9° at $K = 10$) and with training window length (30.0° at $T = 3$yr to 15.8° at $T = 10$yr). Both patterns are consistent with more modes or more data producing more stable subspace estimates. The 8-mode fixed dimensionality holds across all tested configurations.

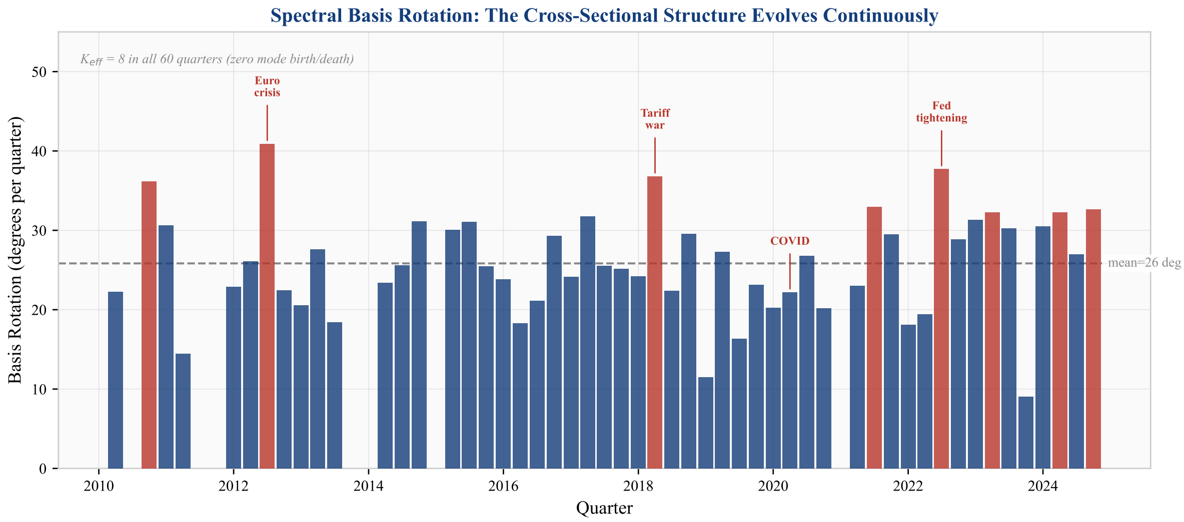

Figure 10: Spectral basis rotation over time (descriptive with bootstrap CIs). Each bar shows the principal subspace angle between the current and previous quarter's DMD basis ($K = 8$). Red bars highlight high-rotation quarters ($> 32°$) aligned with macro events. The dashed line marks the mean ($25.8°$). Null simulation from fixed-basis panels produces $75.3°$ apparent rotation, suggesting the observed values reflect consistently recoverable structure rather than estimation noise.

The standard Kalman filter, operating in a fixed basis U, tracks the modal states $\boldsymbol{\alpha}_t$ but is blind to this basis rotation. Rolling basis update (recomputing U each quarter with the latest data) captures the rotation, yielding modal $R^2 = 0.696$ on the structural panel—a +14.7 percentage point improvement over the static-basis pipeline (CI [+11.1, +17.2]pp from the ablation bootstrap in Table 5) and winning all 10 test windows (Table 8).

An ablation confirms that the improvement is from basis update, not from resetting the Kalman state: resetting the state without updating the basis *hurts* by -3 percentage points, because the accumulated Kalman state carries valuable information that the reset discards.

Configuration	Mean modal R^2	Std	Δ (rolling $-$ static)
SMIM (rolling basis)	**0.696**	0.050	+0.147
SMIM (static basis)	0.549	0.075	—

Table 8: Rolling vs static basis comparison across 10 annual windows (93-actor CapEx/Assets panel, fixed $K=8$, $T=5$yr, modal R^2). Modal R^2 measures spectral reconstruction quality, not forecast accuracy. The headline predictive results (Table 3) use the CapEx/Revenue panel with predictive alpha.

4.11 Cross-Sectional Pooling: Training Window Effect

A distinctive property of the spectral framework is that its forecast quality degrades less than per-actor AR(1) as the training window shortens. Figure 11 shows ΔR^2 (SMIM minus AR(1)) as a function of training window length T, at fixed $K = 2$ and $\tau = 12$Q.

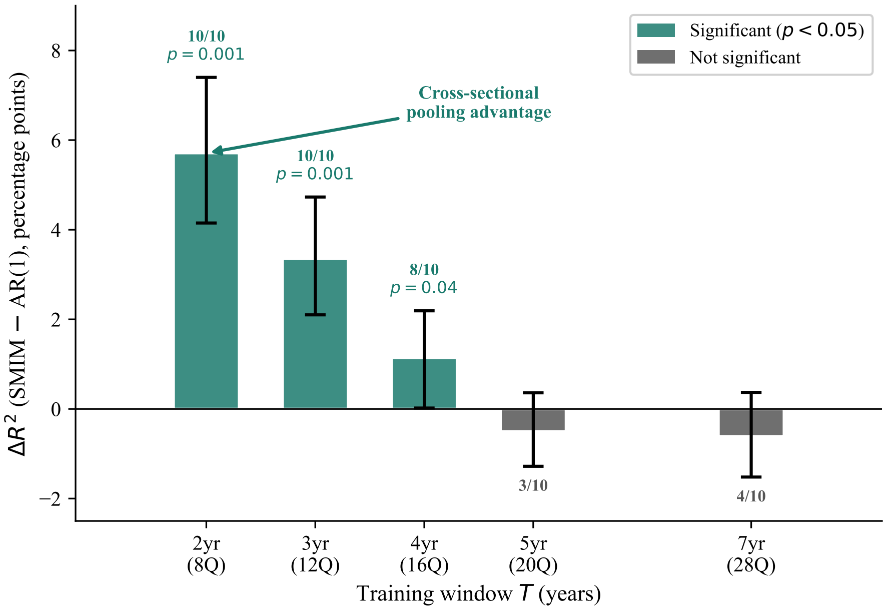

Figure 11: Training window effect on forecast advantage (fixed $K = 2$, $\tau = 12$Q). SMIM's advantage over AR(1) increases monotonically as T decreases. At $T = 2$yr, AR(1) has only 8 quarterly observations per actor to estimate ρ; SMIM extracts its 2-mode spectral basis from $146 \times 8 = 1{,}168$ cross-sectional observations. Error bars show 95% bootstrap CI (10,000 resamples). Teal bars are significant at $p < 0.05$ (permutation test). Win counts show SMIM-vs-AR(1) per-window wins out of 10.

The pattern is consistent with *cross-sectional pooling*: the DMD basis is estimated from the joint $N \times T$ panel, effectively borrowing strength across actors. Per-actor AR(1) has no mechanism for this—each actor's autoregressive coefficient is estimated from T observations alone. As T shrinks, the per-actor estimates degrade while the cross-sectional structure remains recoverable from the N-dimensional snapshots. At $T = 7$ years (28 quarters), AR(1) has sufficient data and SMIM adds nothing; at $T = 2$ years, the cross-sectional advantage produces a +5.7pp gain with bootstrap CI entirely above zero ($p = 0.001$).

Predictive vs modal R^2. The Kalman filter produces both a *predictive* state $\boldsymbol{\alpha}_{t|t-1} = F\boldsymbol{\alpha}_{t-1|t-1}$ (using only information up to $t-1$) and a *filtered* state $\boldsymbol{\alpha}_{t|t}$ (incorporating the current observation $\tilde{y}_t$). All forecast comparisons in this paper use the predictive state; the filtered state is used only for investment gap computation (Section 5.2), where it provides a better reconstruction of the current cross-section.

Structural analysis panel. The ablation ladder (Table 5), basis rotation analysis, variance decomposition, and CapEx revision prediction (Section 5.2) use a broader 93-actor panel with $K = 8$ modes and filtered (modal) R^2. Modal R^2 measures spectral reconstruction quality—how well the DMD basis represents the current cross-section—which is the relevant metric for structural decomposition. These findings (rolling basis dominates, dual regularisation is essential, gaps predict CapEx revision) are structural and do not depend on the forecast comparison metric.

5 Economic Validation

5.1 What This Gap Is and Is Not

The investment gap $\Delta_{i,t}$ is a *model-implied* deviation: the difference between observed investment intensity and a spectral-state-space benchmark (Eq. 9). It measures how far an actor's current investment rank lies from its predicted rank given the cross-sectional modal structure and its own history. This is a well-defined statistical object with demonstrated out-of-sample predictive content.

It is *not*, however, a welfare-theoretic or structural misallocation measure in the sense of Hsieh and Klenow (2009). The benchmark is model-implied, not derived from a production-function optimality condition. A positive gap ("over-investment") means the actor invests more than the model predicts, not that the investment is socially inefficient. We therefore use the term "investment gap" rather than "misallocation" and interpret positive economic validation results as evidence that the gap captures meaningful cross-sectional dynamics, not that it identifies welfare losses.

5.2 Gaps Predict CapEx Revision (Main Validation Result)

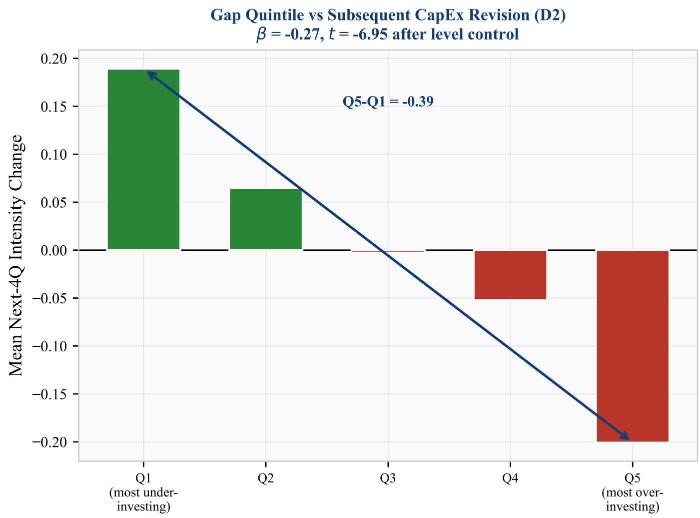

Figure 12: Gap quintile sort (inferential): mean next-4-quarter intensity change by current gap quintile. Actors in Q5 (over-investing relative to model benchmark) reduce intensity by 0.20 rank points; Q1 (under-investing) increase by 0.19. Panel regression with clustered SEs yields $t = -6.4$ to -7.1 depending on specification (Table 9).

The strongest economic validation result is that investment gaps predict subsequent CapEx revision. We estimate the panel regression:

$$\Delta y_{i,t\to t+4} = \beta_0 + \beta_1 \Delta_{i,t} + \beta_2 y_{i,t} + \epsilon_{i,t}. \tag{10}$$

Table 9 reports six specifications. In pooled OLS with the level control: $\hat{\beta}_1 = -0.28$, $t = -6.4$ (actor-clustered SEs), $N = 3{,}324$. The result is robust to time-clustered SEs ($t = -7.1$) and to time fixed effects ($t = -7.1$). Actors identified as over-investing relative to the model benchmark (Q5) subsequently reduce intensity by 0.20 rank points; under-investing actors (Q1) increase by 0.19 (Figure 12).

Important caveat: actor fixed effects. With actor FE (within-actor variation only), $\hat{\beta}_1$ flips sign to $+0.08$ ($t = +2.2$). This means the cross-sectional component—which actor is above or below its model-implied benchmark—drives the main result. *Within* an actor over time, gaps predict slight continuation rather than reversion. The pooled result is therefore a cross-sectional prediction ("actors with large gaps tend to revert"), not a within-actor prediction ("an actor's gap predicts its own next move"). This distinction is important for interpretation.

Specification	$\hat{\beta}_{\text{gap}}$	SE	t	N
Pooled OLS (HC1)	−0.279	0.034	−8.3	3,324
Pooled OLS (actor-clustered)	−0.279	0.044	−6.4	3,324
Pooled OLS (time-clustered)	−0.279	0.040	−7.1	3,324
Time FE (time-clustered)	−0.280	0.040	−7.1	3,324
Actor FE (actor-clustered)	+0.084	0.039	+2.1	3,324
Two-way FE (actor-clustered)	+0.085	0.039	+2.2	3,324

Table 9: CapEx-revision panel regression (inferential). Dependent variable: next-4-quarter intensity change. All specifications include the current intensity level as control. Actor-clustered and time-clustered SEs shown. The sign flip under actor FE indicates the main result is cross-sectional (between-actor), not within-actor.

5.3 Gap Persistence

Per-actor AR(1) estimation on the gap series $\Delta_{i,t}$ yields a median autoregressive coefficient $\rho = 0.67$ and median half-life of 1.7 quarters. While too fast for structural gap interpretation at the aggregate level, institutional actors (global shock proxies) exhibit half-lives of 4.7 quarters. Cross-sectional quintile persistence is strong: 49.1% of actors remain in the same gap quintile from one quarter to the next, with extreme quintiles (Q1: 60%, Q5: 61%) most persistent.

5.4 Exploratory Results

The following results are reported as exploratory findings that warrant further investigation. They are not part of the paper's main claims.

5.4.1 Network Diffusion

Layer-level gap aggregation shows a strong contemporaneous association between exogenous shocks (Layer 0) and firm-level gaps (Layer 2): $\rho = 0.86$ at 1-quarter lag ($p < 0.001$). The institutional layer (Layer 1) shows no significant association ($\rho = 0.03$, $p = 0.83$). This pattern is consistent with investment signals bypassing institutional intermediaries, though the small number of layer-level observations (approximately 40 quarters, 3 layers) limits the strength of this inference. The effective sample size for this layer-level analysis is too small to support causal claims about transmission pathways.

5.4.2 Gap Dispersion and Market Volatility

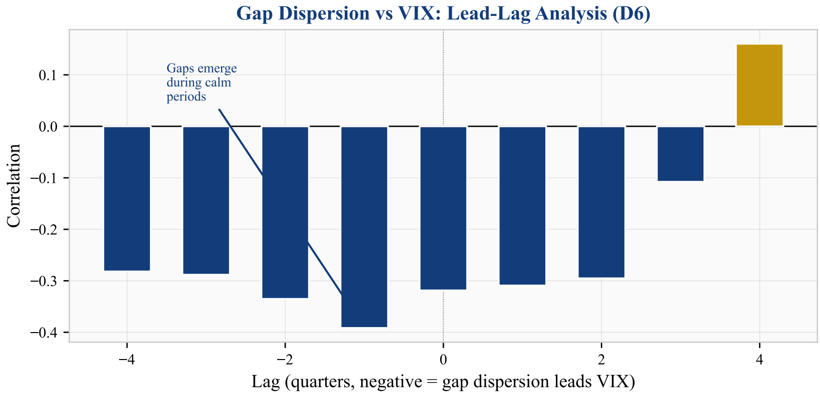

Figure 13: Lead-lag correlation between gap cross-sectional dispersion and VIX (exploratory). Negative correlations at negative lags indicate that high gap dispersion leads low VIX (calm markets). Based on ∼40 quarterly observations with 9 tested lags; would not survive conservative multiple-testing correction (see caveat in text).

Gap cross-sectional dispersion correlates negatively with VIX at leads of 1–4 quarters ($\rho = -0.39$ at 1Q lead; Figure 13). This finding—that investment gaps widen during calm markets—is consistent with the "volatility paradox" hypothesis (Brunnermeier and Sannikov, 2014).

Caveat. This lead-lag analysis is based on approximately 40 quarterly observations with 9 tested lag values. The correlation of -0.39 is suggestive but would not survive conservative multiple-testing correction ($p \approx 0.01$ uncorrected for 9 comparisons). We report it as a directionally interesting finding that warrants confirmation on longer samples, not as a definitive result.

6 Discussion

6.1 What Does Not Work

Emergence diagnostics. On the structural panel, we tested emergence through five distinct channels: PID synergy between spectral modes, TDA topological complexity, economic emergence features (cross-sectional dispersion, sector rotation velocity, Herfindahl concentration), Extended DMD (Koopman lifting with polynomial observables), and multi-resolution DMD divergence. None produces a positive ΔR^2. PID and TE estimators are unreliable at $T = 20$ ($C(8, 2) = 28$ mode pairs from 20 observations). Extended DMD with degree-2 polynomial lifting ($P = 44$ features) achieves modal $R^2 = 0.19$—far worse than linear DMD at 0.54—because $T/P = 0.45$ is underdetermined. Economic emergence features are redundant with the DMD-Kalman pipeline ($\Delta R^2 = -0.003$).

Directed spectral operators. Transfer entropy and Granger causality operators produce genuinely asymmetric matrices (mean asymmetry ~ 1.2). However, the resulting spectral bases are too noisy for Kalman filtering: TE-Schur at $K = 8$ causes numerical divergence, and at $K = 3$ achieves only modal $R^2 = 0.36$ vs DMD's 0.54.

Event-level alignment. Gap magnitudes do not spike around known episodes (0/8 aligned). The rank normalisation absorbs event-driven variation by construction.

Return-based intensity. On the structural panel, return intensity produces modal $R^2 = -0.15$ on the same US actors where CapEx/Assets intensity achieves modal $R^2 = 0.54$. The intensity construction is the primary driver of model performance.

6.2 When the Kalman Filter Is and Is Not Needed

At $K = 2$ with quarterly basis refresh, the Kalman filter's contribution to predictive R^2 is marginal (+0.3pp). The rolling basis update resets the modal state each quarter, leaving only a single predict step—$\boldsymbol{\alpha}_{t|t-1} = 0.99 \cdot U_t^\top \tilde{y}_{t-1}$—that is functionally equivalent to a shrunk basis projection. The full predictive pipeline thus reduces to:

$$\hat{y}_{i,t} = \hat{\mu}_i + 0.99 \cdot U_t U_t^\top \tilde{y}_{i,t-1}, \tag{11}$$

a per-quarter DMD eigendecomposition followed by a matrix-vector product.

The Kalman filter becomes essential in two settings: (i) at higher K, where spherical R prevents overfitting the K-dimensional observation model (the structural panel's $K = 8$ requires it; without spherical R, modal R^2 drops from 0.44 to −7.3); and (ii) when the basis is *not* refreshed each quarter, where online Q adaptation tracks regime shifts between basis updates (+5.8pp on the structural panel). These two roles—regularisation and temporal adaptation—are redundant at $K = 2$ with rolling basis, but load-bearing at higher K or lower update frequency.

This separation clarifies a broader point: the structural panel ($K = 8$) recovers a richer spectral representation that requires Kalman regularisation; the predictive panel ($K = 2$) uses only the two most stable modes, for which a simple projection suffices. The 6 additional modes captured at $K = 8$ represent real cross-sectional structure (they survive the null-simulation test and rotate coherently), but are too noisy to improve one-step-ahead forecasts. Understanding how to exploit these additional modes for prediction is an open question.

6.3 Reading the Spectral Structure: A Practitioner's Perspective

The structural analysis reveals that cross-sectional investment dynamics live in a stable 8-dimensional spectral subspace that rotates continuously. We conclude the discussion by interpreting what this means concretely for someone monitoring capital allocation patterns.

The 8 modes are investment co-movement themes. Each DMD mode $u_k \in \mathbb{R}^N$ assigns a loading to every actor. Actors with same-sign loadings tend to increase or decrease their investment intensity together; opposite-sign loadings indicate substitution patterns. These are not fixed sectors—they are *empirically identified* co-movement directions that cut across sector boundaries.

26°/quarter rotation means the themes drift. Every quarter, approximately 10% of the co-movement structure reshuffles: firms that invested in sync gradually decouple, while previously independent firms begin to co-move. After one year (∼100° cumulative rotation), the investment map differs substantially from where it started. After three years, the spectral directions are nearly orthogonal to their initial orientation. Any model of cross-sectional investment dynamics that is not re-estimated at least quarterly will track a structure that no longer exists.

Rotation speed is diagnostic. During business-as-usual quarters, the basis rotates at 26° ($\sigma = 6.5°$). High-rotation quarters ($> 35°$) signal structural reorganisation: the tariff war (2018 Q2, 37°) did not merely change *how much* firms invested—it changed *which firms co-moved.* The European debt crisis (2012 Q3, 41°) and the Federal Reserve's tightening cycle (2022 Q3, 38°) similarly reorganised the cross-sectional pattern. This is a structural signal distinct from volatility: VIX measures how much prices move; rotation measures how the investment co-movement map rearranges.

Stable dimensionality with shifting directions. The number of independent co-movement themes ($K_{\text{eff}} = 8$) is constant across all 52 quarterly transitions in our sample. No modes are born or die. The investment structure does not become simpler during crises or more complex during expansions—the complexity is fixed, but the directions evolve. This is consistent with a system that has a fixed number of structural degrees of freedom (sector composition, supply-chain linkages, policy transmission channels) whose relative importance shifts gradually.

Practical implications. (i) Factor models for investment-related portfolios (infrastructure, CapEx-intensive sectors) must be updated at least quarterly; annual re-estimation loses $\sim 100°$ of rotation. (ii) Rotation speed can serve as an early diagnostic for structural regime change—a sustained increase in rotation may signal that the investment landscape is reorganising before this becomes visible in asset prices. (iii) The 8-dimensional structure provides a natural compression: 146 firms' investment behaviour can be summarised by 8 time-varying modal coordinates per quarter, enabling compact monitoring dashboards. (iv) Of the 8 recoverable modes, only 2 carry enough signal for one-step-ahead prediction (Section 4.1); the remaining 6 capture real structure that is visible in reconstruction but not yet exploitable for forecasting—an open frontier for future work.

6.4 Limitations

Both panels (93-actor structural and 146-firm headline) are US-dominated, with limited international evidence. UK results are weak (modal $R^2 = 0.058$ on the structural panel) due to the return-based intensity method, not geography per se. Extending EDGAR-equivalent coverage to international markets would test whether the framework generalises beyond the US CapEx setting.

The 4-quarter test windows provide noisy R^2 estimates per window. Window-level inference with 8–10 observations has limited power; we address this with complementary actor-quarter-level Diebold–Mariano tests (n = 4,600, Newey–West HAC) and the training-window sweep (Figure 11), which demonstrates a consistent monotonic pattern across five T values.

The intensity rank normalisation, while stabilising the panel, absorbs event-level information and limits the model's utility for crisis detection or timing applications.

7 Conclusion

We present a spectral state-space framework for forecasting cross-sectional investment intensity. On a 146-firm US CapEx/Revenue panel, a 2-mode DMD basis with dual-regularised Kalman filtering achieves predictive $R^2 = 0.711$ under nested CV and $R^2 = 0.845$ on frozen holdout, exceeding per-actor AR(1) at the actor-quarter level (Diebold–Mariano $p < 0.001$). The advantage grows monotonically as the training window shrinks, reaching $\Delta R^2 = +0.057$ at $T = 2$ years (10/10 windows)—a cross-sectional pooling effect where the spectral basis leverages all $N \times T$ observations while AR(1) uses each actor's T alone. The result rests on three principles:

Dual regularisation: both the observation covariance ($R = cI$, replacing N^2 parameters) and the transition matrix ($F = 0.99I$, replacing K^2 parameters) must be aggressively regularised. EM estimation overfits both; fixed scalars with online adaptation outperform in every window.

Temporal basis tracking: on the structural panel, the spectral basis rotates at approximately 26° per quarter with stable dimensionality. This rotation, missed by a static basis, accounts for the single largest modal R^2 gain (+14.7pp, CI $[+11.1, +17.2]$pp). Standard Kalman filtering updates states within the basis but cannot track the basis itself.

Systematic ablation with uncertainty: each pipeline component is ablated with bootstrap confidence intervals across 10 windows. The largest gains come from eliminating failure modes: spherical R, transition regularisation, and basis tracking—all with CIs entirely above zero (Table 5).

The model-implied gap estimates predict subsequent CapEx revision ($t \approx -6.4$ to -7.1 with clustered SEs), though this is a cross-sectional result that flips sign under actor fixed effects (Table 9). The transition dynamics use stable default scalars ($F = 0.99I$, $Q = 0.5I$) that performed consistently across all evaluation windows in this sample—only the spectral basis requires periodic re-estimation.

Cross-sectional pooling. The framework's advantage increases monotonically as the training window shortens, from near-zero at $T = 7$ years to $\Delta R^2 = +0.057$ at $T = 2$ years (Figure 11). This pattern reflects the spectral basis exploiting $N \times T$ cross-sectional observations collectively, while per-actor AR(1) is limited to each actor's own history. The effect is strongest precisely when per-actor data is most scarce.

Scope conditions. The headline predictive result is on a 146-firm US CapEx/Revenue panel. The structural findings (ablation, rotation, economic validation) are established on a broader 93-actor CapEx/Assets panel using modal (filtered) reconstruction. Return-based intensity (UK equities) and asset-growth intensity (banks) produce weak or negative R^2. Generalisation to other intensity constructions, geographies, and data frequencies requires further work. The methodological lesson—dual regularisation and basis tracking for high-dimensional state-space models—extends beyond this specific application.

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Disclaimer. This paper presents a research framework for academic investigation of investment dynamics. It is not investment advice. The models, gap estimates, and statistical findings described herein have not been validated for live trading or portfolio construction. Past out-of-sample performance on historical data does not guarantee future results. The author bears no responsibility for any financial losses incurred by parties who deploy this framework or its derivatives in investment strategies. Any such deployment is undertaken entirely at the user's own risk and should be subject to independent due diligence, regulatory compliance review, and appropriate risk management.

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